



Antimicrobial Stewardship Policy

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All services	Pharmacy, Medical and Nursing

2. Purpose

This policy establishes key directives for antimicrobial prescribing and management at Armada Kalamunda Group (AKG), with reference to antimicrobial prescribing processes, management of antimicrobial formulary and restrictions, use of evidence-based therapy and access to expert clinical advice.

3. Background

Antimicrobial stewardship (AMS) can be defined as an ongoing and systematic effort to optimise the use of antimicrobial medicines within a health service organisation. The key objectives of AMS programs include reducing inappropriate antimicrobial use, improving patient care outcomes, and mitigating adverse consequences of antimicrobial use (such as antimicrobial resistance, preventable patient harm and unnecessary costs associated with pharmaceutical expenses and drug-resistant infections).

4. Principles of antimicrobial prescribing

4.1 Key principles for judicious antimicrobial use

All prescribers across the AKG are expected to prescribe antimicrobial therapy according to the following key principles outlined in the Antimicrobial Stewardship Clinical Care Standard (2020) Quality Standards:

- Therapeutic decisions regarding the prescription of antimicrobials will be based on best available evidence.
 - Empirical antimicrobial therapy (the infective pathogen is not known) or prophylactic therapy (given to prevent acquisition or development of infection) is prescribed according to locally endorsed guidelines or Therapeutic Guidelines: Antibiotic.
 - When the infective pathogen is known, antimicrobials are to be prescribed according to microbiology results and antimicrobial susceptibilities, where available. Investigation results are reviewed promptly when they are reported.
- A patient with a life-threatening condition due to a suspected infection receives an appropriate antimicrobial immediately, without waiting for the results of investigations
- Prescribing antimicrobials will be of the narrowest spectrum possible for achieving the intended effect.

- When a patient is prescribed an antimicrobial, the indication, active ingredient, dose, frequency and route of administration, and the intended duration or review plan are documented in the patient's healthcare record. Thorough documentation of adverse reactions to antimicrobials as per the 2020 Antimicrobial Stewardship Clinical Care Standard (see [Adverse Drug Reaction and Clinical Alert Procedure](#))
- A patient having surgery, or a procedure is prescribed antimicrobial prophylaxis in accordance with the current Therapeutic Guidelines or evidence-based, locally endorsed guidelines. This includes recommendations about the need for prophylaxis, choice of antimicrobial, dose, route and timing of administration, and duration. For example, most surgical prophylaxis should NOT be prescribed for > 24 hours. (See [Surgical Prophylaxis Guideline](#))

Summary of principles of judicious antimicrobial use:

- M** – Microbiology guides therapy (where possible)
- I** – Indications should be evidence based
- N** – Narrowest spectrum required
- D** – Dosage appropriate to the site and type of infection
- M** – Minimise duration of therapy
- E** – Ensure monotherapy (wherever possible)

4.2 Use of evidence-based prescribing guidelines

- Locally endorsed guidelines

All locally endorsed antimicrobial prescribing guidelines are guidelines that have been reviewed and approved by the AKG AMS Committee that reports to the AKG Drug and Therapeutics Committee and should be consistent with the Antibiotic eTG. Any deviations to the Antibiotic eTG should be approved by the AMS committee as above.

Children's Antimicrobial Stewardship Program (ChAMP) Guidelines

The AMS Committee supports the use of the [ChAMP Clinical Guidelines](#) for reference for clinical support of antimicrobial use for paediatric patients at Armadale Kalamunda Group.

- Therapeutic Guidelines: Antibiotic

The current version of [Therapeutic Guidelines: Antibiotic](#) should be used in clinical scenarios for which a locally-endorsed antimicrobial prescribing guideline does not exist. These guidelines are an Australian resource developed by a multidisciplinary expert writing group, with recommendations based on best available evidence for empirical efficacy, harm minimisation and other important considerations such as antimicrobial resistance and individual patient factors.

- Prescribing outside locally endorsed guidelines or Therapeutic Guidelines: Antibiotic

If a patient is prescribed antimicrobial therapy that is not in concordance with either locally-endorsed guidelines or [Therapeutic Guidelines: Antibiotic](#), the prescriber is expected to document their clinical reasoning in the patient's medical record.

Where an indication or clinical scenario is not covered by either set of guidelines, a prescriber may choose antimicrobial therapy based on an appropriate published peer-reviewed source. Care should be taken to ensure the recommendations made are relevant to an Australian context in terms of both antimicrobial resistance patterns and availability of the selected antimicrobial agents. Alternatively, a prescriber may wish to seek expert advice via an Infectious Diseases/Clinical Microbiology consultation.

4.3 Parenteral to oral conversion

A switch from intravenous (IV) administration to the oral formulation of an antimicrobial is to be made as soon as it is safe to do so and IV to oral switch is to be promoted by pharmacists, clinicians, and the AMS Committee.

The criteria for parenteral to oral conversion may include the following (only one of the criteria is required at a minimum for the consideration of oral conversion):

- When the patient is tolerating oral fluids/food and there is no reason to believe poor oral absorption from the gastrointestinal tract
- Temperature is less than 38°C or improving over 24 hours
- Signs and symptoms of infection are improved or resolved
- An appropriate oral alternative is available.

All patients on antimicrobial agents are to be reviewed at a minimum at each treating team ward round to consider whether it is clinically appropriate to de-escalate to a narrower spectrum agent, to switch from IV to oral agents, or to cease antimicrobials.

5. Access to prescribing resources and support

5.1 Access to evidence-based prescribing guidelines

- [Therapeutic Guidelines: Antibiotic](#) can be accessed through eTG complete via the Library Services.

Antibiotic approval codes are given through the electronic prescribing program or Infectious Diseases/Microbiology Clinical Service

5.2 Access to antimicrobial advice

- Prescribers at AKG have access to expert advice on antimicrobial prescribing and the management of infectious diseases 24 hours a day, seven days a week.
- During business hours (Mon-Fri, 8:30am-4:30pm), prescribers should contact the Infectious Diseases Department or Clinical Microbiology, via the AHS switchboard (08) 9391 2000
- After hours, prescribers may contact the on-call medical officer for Infectious Diseases or Clinical Microbiology, via the RPH switchboard (08) 9224 2244.

Paediatric and neonatal patients' antimicrobial advice:

For antimicrobial advice for **neonate and paediatric** patients – staff can contact the Infectious Diseases Department at Perth Children's Hospital (PCH) via the PCH switchboard (08) 6456 2222.

The treating team is the first point of call for general advice regarding antimicrobial prescribing and approval of restricted antimicrobial agents. In more complex cases, the treating team may recommend a consultation by an Infectious Diseases Physician.

6. Local restrictions on antimicrobial prescribing

6.1 Formulary and non-formulary antimicrobials

A list of antimicrobial formulary items is available via [Formulary One](#) (WA State-wide Medicines Formulary).

- Many antibiotics have restrictions or conditions on their use as part of efforts to prevent the development of drug resistance (see 6.2)
- These antibiotics are termed “restricted”
- Restricted anti-microbials are to be stored in marked storage areas on the wards
- Non-formulary antimicrobials are not routinely stocked by AKG. The use of a non-formulary product for an individual patient requires application to and approval by the Director Clinical Services; however, this process may be expedited for urgent antimicrobial therapy.
- Requests for non-formulary antimicrobials will usually require consultation by an Infectious Diseases Consultant prior to approval, or as a condition of approval.

6.2 Antimicrobial restrictions

The implementation and maintenance of antimicrobial restrictions is a core AMS strategy at AKG. The restriction status and access rules for each antimicrobial agent (or specific formulations) are based on the following considerations:

- Best practice prescribing.
- Patient safety risks associated with antimicrobials (including adverse effects, drug interactions, the need for therapeutic drug monitoring, risk of *Clostridioides difficile* infection and risk of selecting for more resistant organisms).
- Complexity of relevant disease states and/or public health risks.
- Potential for contributing to antimicrobial resistance in the healthcare environment.
- Cost-effectiveness (as compared to alternative agents).
- Resources available to manage antimicrobial restrictions.

Although [Formulary One](#) lists antimicrobial availability, restrictions, and Pharmaceutical Benefits Scheme (PBS) indications, there may be additional requirements for certain antimicrobial prescribing at AKG due to local issues such as drug supply.

Antibiotic approval codes are given through the electronic prescribing program or Infectious Diseases/Microbiology Clinical service

Please see Appendix 1 for how to prescribe restricted antimicrobials

7. Conflict resolution

7.1 Disagreements regarding restricted antimicrobials

Where a request for a restricted or highly restricted antimicrobial has been denied, the Infectious Diseases Physician/Clinical Microbiologist will attempt to work collaboratively with the relevant prescriber or medical team to come to a mutually agreeable therapeutic decision.

If no agreement can be reached, the medical practitioner responsible for the patient may pursue prescribing the restricted or highly restricted antimicrobial in an individual patient case, only on the grounds of a perceived risk to patient safety.

This means:

- Without use of the restricted or highly restricted antimicrobial, the medical practitioner considers the patient to be at significant risk of harm, and
- The medical practitioner believes that use of the restricted or highly restricted antimicrobial will not put the patient at an equivalent or higher risk of harm, and
- The medical practitioner believes that alternative therapies will not have a superior risk-benefit profile in this patient.

A decision to prescribe the requested restricted or highly restricted antimicrobial agent, despite contrary advice from Infectious Diseases Department (IDD)/Clinical Microbiology, must be communicated to IDD/Clinical Microbiology directly and documented in the patient's medical record.

Cases where restricted or highly restricted antimicrobials are prescribed against the advice of IDD/Clinical Microbiology, will be tabled, and reviewed by the AMS Committee at the next scheduled meeting, to address concerns and inform future initiatives such as prescriber education campaigns or quality of prescribing audits.

7.2 Supply of antimicrobials against the advice of the AMS team

Where restricted or highly restricted antimicrobials are prescribed against the advice of IDD/Clinical Microbiology, a pharmacist may supply the antimicrobial providing:

- The attending medical officer is choosing to prescribe against the advice of IDD/Clinical Microbiology on the grounds of a perceived risk to patient safety (see the [Disagreements regarding restricted antimicrobials](#) section); and
- The attending medical officer (or their delegate) has documented their decision and reasoning in the patient's medical record; and
- IDD/Clinical Microbiology has been informed (or will be informed on the next business day); and
- The pharmacist does not have any patient safety concerns about use of the antimicrobial as prescribed (i.e., dose is safe, nil contraindications, nil clinically significant drug-drug interactions, route of administration is appropriate).

For high cost, highly restricted antimicrobials the above conditions must be met and where possible, the pharmacist involved must also discuss supply with the Head of Department Pharmacy, who may escalate any matters for consideration to the Director Clinical Services for resolution.

Patients should never have their anti-microbials withheld because there is no approval code.

8. Compliance, monitoring and evaluation

The AMS Sub-Committee will be accountable for monitoring compliance with this policy.

Compliance will be monitored via (at minimum):

- Pathology data provided by Path West Laboratory in the form of a standard antibiogram to be tabled at the AMS Committee on an annual basis.
- Antimicrobial usage data via participation in the National Antimicrobial Utilisation Surveillance Program (NAUSP) is to be regularly analysed by the AMS committee to provide antimicrobial usage data (ideally normalised to usage density rates) and usage reports of antimicrobial agents.
- Annual completion of a National Antimicrobial Prescribing Survey (NAPS) and Surgical National Antimicrobial Prescribing Survey (SNAPS).
- Relevant data (i.e., audits, education sessions, completion of online learning modules, and quality use of medication indicators) should be regularly monitored to identify local risks, inform the focus of local programs, and evaluate the effectiveness of the local antimicrobial stewardship program activities.

It is the responsibility of the AMS sub-committee to ensure any significant risks, results, and recommendations of the above monitoring to be disseminated to the relevant clinical teams in a timely manner

Other quality activities (e.g., audits) are encouraged and their results should be used to identify opportunities for improvement.

9. Roles and responsibilities

Role	Responsibility
Antimicrobial Stewardship Committee	• Define restriction status and approval process for antimicrobial agents prescribed within the hospital and health service. At a minimum, the approval process should include a discussion with an Infectious Diseases Physician / Clinical Microbiologist that is documented in the patient's medical record and is therefore auditable. • Review and recommend any alterations to restrictions, audit antimicrobial use and oversee related education. • Ensure medication safety is a key consideration in all formulary decisions and antimicrobials considered to be high-risk are included in the high-risk medication list when added to the formulary. • Development and endorsement of policies, protocols and guidelines relating to antimicrobial use.

Role	Responsibility
	• Monitor antimicrobial use and analyse antimicrobial resistance reported by the pathology service and policy compliance rates and formulate corrective action, if indicated. • Please refer to the terms of reference of the AMS committee
Authorised Prescribers	• Must comply with the principles of good antimicrobial prescribing in this policy • Must document the indication and expected duration for prescription of all antimicrobials in the medical record, along with a review plan; and • Must comply with the local approval process for restricted antimicrobials, as well as manage any therapeutic drug monitoring if required. • Ensure that a patient with a suspected infection, and/or their carer, receives information on their health condition and treatment options in a format and language that they can understand. • Ensure that when a patient is prescribed antimicrobials, information about when, how and for how long to take them, as well as potential side effects and a review plan, is discussed with the patient and/or their carer. This includes providing information on the patient's health condition/infection and the treatment options available.
Pharmacists	• Are responsible for reviewing antimicrobial orders for adherence to local (or Therapeutic) guidelines; and • Providing timely feedback, where applicable, to the prescriber. • When therapeutic drug monitoring is necessary, the pharmacist advises on sampling time, and assists in interpretation of results and adjustment of dosing regimen.
Midwives and Nurses	• Ensure antimicrobials are administered in accordance to the prescription • Ensure six rights of medication administration are met each time, as specified in the Medication Management Procedure.

10. Legislative context

The following documents are required to give effect to this policy:

- *Medicines and Poisons Act 2014*
- *Medicines and Poisons Regulations 2016*
- Health Department of Western Australia (HDWA) [Statewide Medicines Formulary Policy MP 0077/18](#)

11. Supporting information

The following documents support the implementation of this policy:

- [Medication Management Policy AKG-POL-0253-13](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)

12. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.18 Antimicrobial stewardship
- The health service organisation has an antimicrobial stewardship program that:
 - Includes an antimicrobial stewardship policy
 - Provides access to, and promotes the use of, current evidence-based Australian therapeutic guidelines and resources on antimicrobial prescribing
 - Has an antimicrobial formulary that includes restriction rules and approval processes
 - Incorporates core elements, recommendations, and principles from the current Antimicrobial Stewardship Clinical Care Standard¹¹⁷
- Standard 3 – Action 3.19
- The antimicrobial stewardship program will:
 - Review antimicrobial prescribing and use
 - Use surveillance data on antimicrobial resistance and use to support appropriate prescribing
 - Evaluate performance of the program, identify areas for improvement, and take action to improve the appropriateness of antimicrobial prescribing and use
 - Report to clinicians and the governing body regarding:
 - compliance with the antimicrobial stewardship policy
 - antimicrobial use and resistance
 - appropriateness of prescribing and compliance with current evidence-based Australian
 - therapeutic guidelines or resources on antimicrobial prescribing

13. References

1. MacDougall C, Polk R. Antimicrobial stewardship programs in health care systems. *Clinical Microbiology Review* 2005; 18(4):638-656

2. Duguid M, Cruickshank M (eds). 2010. Antimicrobial stewardship in Australian hospitals. Australian Commission on Safety and Quality in Healthcare: Sydney
3. Australian Commission on Safety and Quality in Health Care. [Antimicrobial Stewardship Clinical Care Standard](#). Sydney: ACSQHC, 2014.

14. Glossary

The following definitions are relevant to this policy:

Term	Definition
Antimicrobial Stewardship (AMS)	An ongoing effort by a health service to optimise antimicrobial use in order to improve patient outcomes, ensure cost-effective therapy and reduce adverse sequelae of antimicrobial use, including antimicrobial resistance.
Antimicrobial Formulary	The range of antimicrobial products available for use in a healthcare facility and configured to provide treatment options for the large majority of patient encounters.
Carers	People who provide unpaid care and support to family members and friends who have a disease, disability, mental illness, chronic condition, terminal illness or general frailty. Carers include parents and guardians caring for children or older persons.

15. Policy owner

Enquiries relating to this policy may be directed to:

Title: Infectious Diseases Consultant
 Department: Infectious Diseases AHS
 Email: AMS.AHS@health.wa.gov.au

16. Review

This policy will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 2.5	25/08/2023	25/08/2026	Minor amendment

17. Authorisation

This policy has been authorised and issued by the Director Clinical Service.

Authorised by	Dr Alison Parr – Director Clinical Services
Authorisation date	14/12/2023
Published date	18/12/2023

Appendix 1: Antimicrobial Prescribing (including conflict resolution) Pathway

